

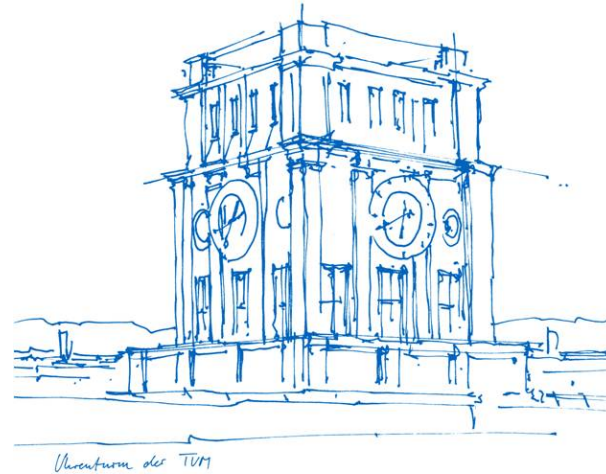


FineFlow–Espresso: A Calibratable 1D Model of Fines Migration

**A continuum model linking fines transport
and hydraulic resistance**

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Presenter transcript

Today I will introduce FineFlow-Espresso as a one-dimensional continuum model of fines migration. The aim is to connect changes in the coffee puck with pressure, flow and fines leaving the puck. The current results demonstrate the numerical model. They are not a calibrated prediction of the experiments. Pore-network modeling is a possible next step for determining closures from the pore geometry.

Presentation outline

Motivation and research question

Physical mechanisms and model scope

Governing equations and closures

Numerical solution and verification

Illustrative results and model sensitivity

Calibration and validation strategy

Conclusions and next steps



Presenter transcript

The presentation has seven sections. We begin with the question and physical mechanisms, then define the governing equations. The numerical section separates algebraic Darcy hydraulics from finite-volume fines transport. Results motivate the subsequent calibration and validation strategy.



Section 1 of 7

Motivation and research question

How do fines change resistance and cup washout?

Presenter transcript

Section 1 of 7: Motivation and research question. How do fines change resistance and cup washout?.

Motivation and research question



Can a 1D model reproduce the experimental response?

- Predict the outlet flow rate for the measured pressure drop.
- Predict the amount of fines reaching the cup.

How does fines redistribution change the hydraulic resistance and cup washout?

Darcy relates pressure to flow through permeability. Deposits change pore space and resistance. Release, capture and remobilization determine the fines distribution.

Presenter transcript

Our primary research question is whether a one-dimensional saturated-puck model reproduces outlet flow for the measured pressure drop and predicts fines collected in the cup. Darcy requires permeability. The structural closures relate permeability to deposits or porosity, while release, capture and remobilization determine the fines inventories. We validate the puck separately from uncertain machine dynamics.



Section 2 of 7

Physical mechanisms and model scope

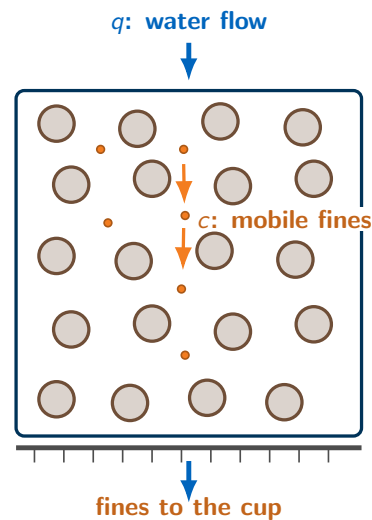
Release, capture and remobilization inside the puck

Presenter transcript

Section 2 of 7: Physical mechanisms and model scope.
Release, capture and remobilization inside the puck.



Flow and fines inside the coffee bed



Flow and mobile fines

- Q : flow rate; A : puck area; $q = Q/A$: superficial velocity;
- c : mobile-fines concentration; $q c$: advective fines flux.

Pore structure and retained fines

- ε : porosity; K : permeability;
- m_a : remaining initially attached fines;
 m_d : recaptured deposits;
- m_a, m_d : mass per bulk volume; c : mass per water volume.

Distinct retained populations assume different kinetics or hydraulic effects.

Presenter transcript

The key distinction is the history of the retained fines. The variable m_a is the remaining releasable inventory that was initially attached to the coffee matrix. It is an evolving state, not just an initial condition. The variable m_d represents fines captured after entering the mobile phase. The current model assigns different release laws to these populations and associates additional blockage with m_d . This is a modeling hypothesis. If experiments do not support the distinction, we should combine them into one retained inventory.



Scope of the 1D continuum model

Present approach

- Forward flow with cross-section-averaged fields varying in z and t .
- CT stage snapshots can constrain axial structural changes.
- Pressure and flow histories constrain the evolving total resistance.

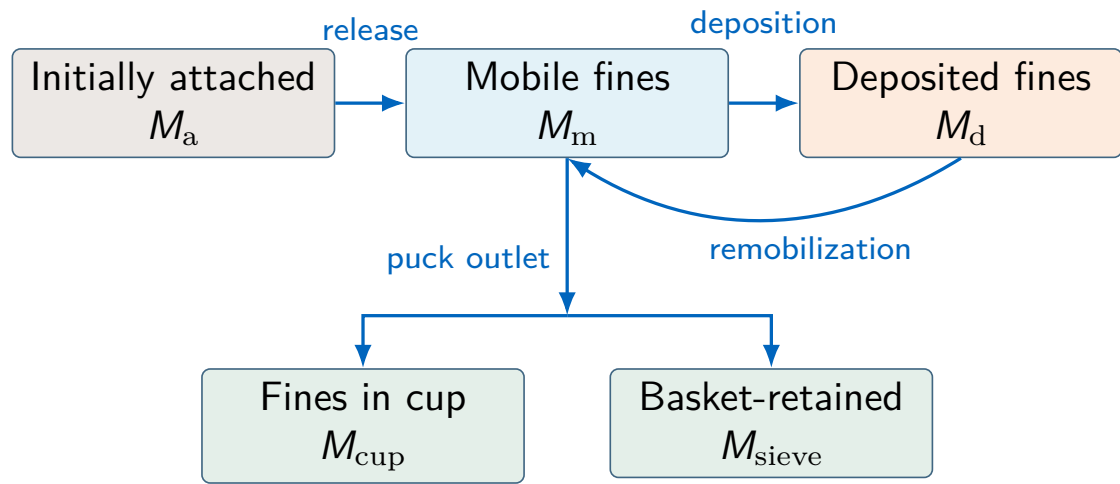
Limits and next measurements

- **Reverse flow and radial channeling are outside the scope of this work.**
- CT shows radial heterogeneity. Measurements must test whether axial averaging captures the overall hydraulic response.
- Time-resolved cup-fines data are a proposed measurement; availability is unconfirmed.
- Pore networks and 2D/3D continuum models are future extensions.

Presenter transcript

The present model considers forward flow and axial variations of cross-section-averaged fields. Reverse flow and radial channeling are explicitly outside the scope of this work. This does not mean radial structure is absent: Mahima's CT observations include radial heterogeneity and blocking clusters. We must compare with measurements to determine whether axial averaging adequately captures the overall hydraulic response. CT images at selected stages are snapshots, not continuous histories. Time-resolved cup-fines measurements remain a proposed input with unconfirmed availability. Pore-network and multidimensional continuum models are possible future extensions.

Fines exchange locally and leave through the outlet



$$M_0 = M_a + M_m + M_d + M_{\text{sieve}} + M_{\text{cup}}.$$

$$M_a = A \int m_a \, dz, \, M_m = A \int \varepsilon c \, dz, \, M_d = A \int m_d \, dz.$$

Accounting domain: puck + basket + collected cup fines.



Presenter transcript

The global balance refers to a specific accounting domain: the entire puck, the fines retained by the basket, and the collected fines in the cup. Capital M denotes actual mass. In particular, mobile mass is the integral of epsilon times c over the puck; epsilon times c alone is a mass density. Local transfers cancel when the inventories are added. With clean inlet water and no creation or uncounted loss of fines, the total equals the initial fines mass.



Section 3 of 7

Governing equations and closures

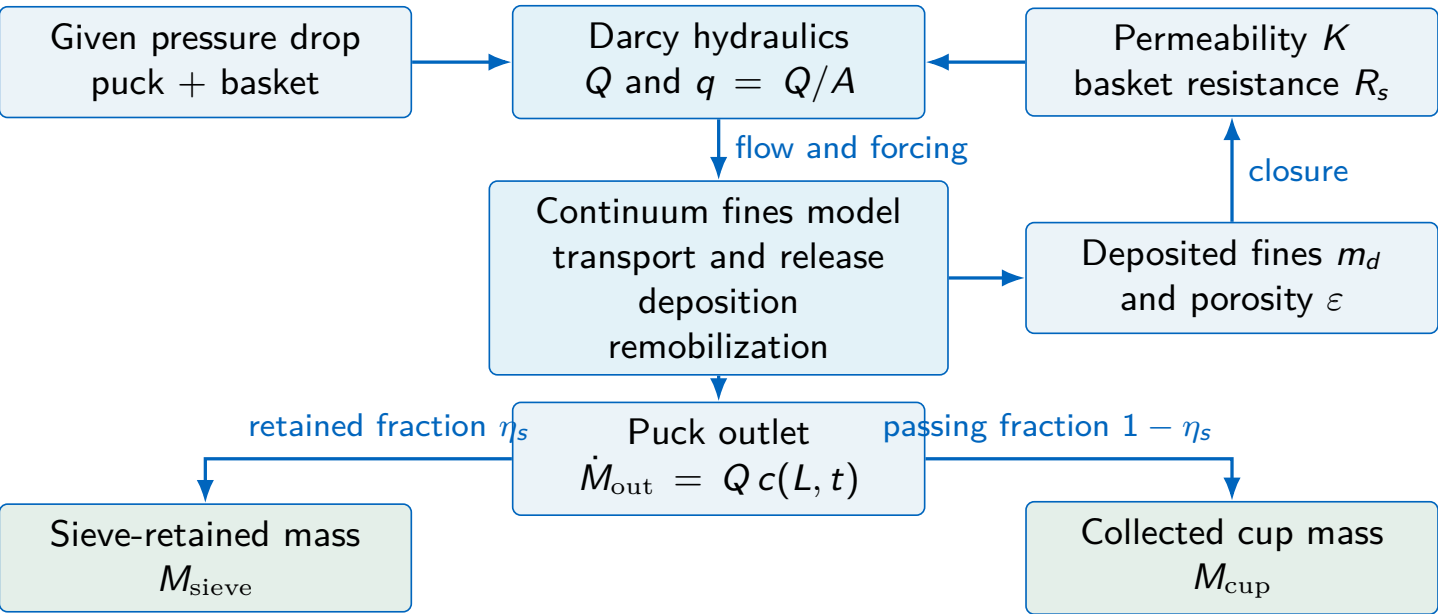
Balances and constitutive choices for fines and flow

Presenter transcript

Section 3 of 7: Governing equations and closures.
Balances and constitutive choices for fines and flow.



Model overview: pressure, flow and fines reaching the cup



Kozeny–Carman uses $\varepsilon(m_d)$ to obtain K . The exponential alternative uses m_d directly.

Presenter transcript

This overview gives the route through Section 3. We want outlet flow for a given pressure drop and the fines collected in the cup. Darcy computes flow using puck permeability and basket resistance. The fines continuum model uses that flow and pressure-gradient forcing to transport fines and update release, deposition and remobilization. Deposits change porosity. Kozeny-Carman obtains permeability from that porosity, whereas the exponential alternative uses deposited mass directly. The loop closes because the new permeability changes flow. At the outlet, flow times the local mobile concentration gives the outgoing fines mass rate. The sieve retains a fraction and the rest reaches the cup. Integrating the rates gives collected masses. The arrows show model dependencies rather than a numerical time-stepping algorithm. The current porosity law responds directly only to deposited fines. Pressure may be prescribed or supplied by the machine model discussed later.



Darcy flow and the permeability closure have different roles

Darcy links the measured pressure drop to outlet flow

$$-\frac{\partial p}{\partial z} = \frac{\mu q}{K(z, t)}, \quad q = Q/A.$$

Lower permeability K means greater resistance and less flow at the same pressure drop.

$$R_{\text{tot}} = \frac{\mu}{A} \int_0^L \frac{dz}{K(z, t)} + R_s, \quad Q = \frac{\Delta p_{\text{total}}}{R_{\text{tot}}}.$$

R_s : basket hydraulic resistance [Pa s m^{-3}]. μ : water viscosity. Δp_{total} : pressure drop across puck plus basket.

Next: how do deposits change K ?

The permeability closure describes that structural response. Fines balances then determine how the deposits evolve.

Presenter transcript

Darcy comes first because flow is one of our two prediction targets. The current permeability determines resistance. Integrating the local Darcy law through the puck and adding basket resistance gives the total pressure-to-flow relation. R_s is the basket hydraulic resistance, with units Pa s per cubic metre. We must match the experimental pressure sensor endpoints. The next slides explain how deposits affect permeability, followed by the fines balances and kinetic closures.



Two alternative closures for the same permeability

Both closures supply K to Darcy. Deposits reduce the available pore space.

$$\varepsilon = \max(\varepsilon_0 - m_d/\rho_d, \varepsilon_{\min})$$

ρ_d : deposit density. Capture raises m_d and remobilization lowers it. Initial release affects porosity indirectly through later capture in this model.

Kozeny–Carman: structural baseline

$$\frac{K}{K_0} = \left(\frac{\varepsilon}{\varepsilon_0}\right)^3 \left(\frac{1 - \varepsilon_0}{1 - \varepsilon}\right)^2$$

Relates permeability to porosity under fixed effective particle scale and Kozeny factor.

This simplified relation may miss changes in pore connectivity and localized plugging.

Exponential: empirical comparison

$$\frac{K}{K_0} = \exp\left(-\beta \frac{m_d}{m_{\text{cap}}}\right)$$

Assumes a direct sensitivity to deposit loading. β controls the imposed blockage strength.

This empirical law has no coffee-data calibration here.

Identical initial K_0 . Neither closure has experimental validation.

Presenter transcript

These are alternative closures for the same permeability, and either can be used with Darcy flow. Kozeny-Carman is our structural baseline: the implemented ratio assumes fixed effective particle scale and Kozeny factor while porosity changes. The exponential relation instead assigns permeability directly from deposited mass. It was introduced as an empirical modeling assumption, not derived from our coffee measurements. Our comparison keeps initial permeability identical, so it tests the evolving relative permeability rather than independently predicting the initial permeability. Neither closure has been validated against the experiments. The displayed exponential results are retained as a sensitivity comparison. In the implemented porosity law, rho_d is deposit density. Only the recaptured deposit inventory directly changes porosity: capture reduces it and remobilization restores it. Initial attached-fines release has no direct pore-volume term. The exponential closure uses deposited mass directly, whereas Kozeny-Carman uses the deposit-dependent porosity.



Why the illustrative model used an exponential law

Motivation: represent strong blockage with a simple adjustable law

The hypothesis was that fines obstruct narrow flow paths, allowing a large resistance increase with a modest change in total porosity.

The exponential form stays positive and decreases smoothly. These useful mathematical properties do not establish physical validity.

m_d/m_{cap}	K/K_0 at $\beta = 5.5$
0	1
0.5	0.064
1	0.0041

An illustrative assumption

Neither the exponential law nor $\beta = 5.5$ was fitted to the coffee data or taken from a cited coffee study. Values shown are before any numerical lower bound.

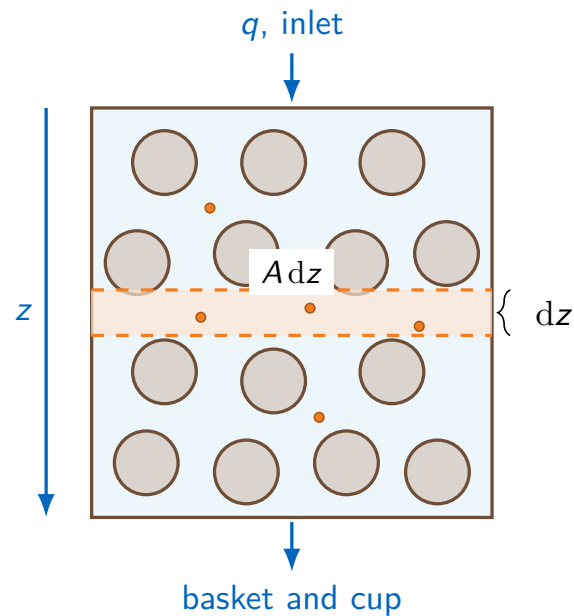
Strong blockage is substantially imposed by this closure; it is not independent evidence of the mechanism.

Presenter transcript

The motivation for the exponential law was to allow strong resistance growth when fines obstruct narrow flow paths, even if the total void volume changes relatively little. That is a physical hypothesis, but the exponential equation itself does not resolve those paths. It was chosen for a simple, smooth, positive response with an adjustable coefficient. The coefficient 5.5 was illustrative and not fitted to our data or taken from a cited coffee study. At half the deposit capacity, the formula already reduces permeability to about 6.4 percent of its initial value. At full capacity it gives about 0.41 percent before numerical bounds. Thus the strong blockage in the original result is substantially imposed by the chosen closure. We should treat Kozeny-Carman as the structural baseline and the exponential law as an empirical sensitivity case until measurements can justify it.



Continuum balance over an infinitesimal slice



Water-filled pore space

$$dM_m = \varepsilon c A dz$$

Mobile fines cross the slice boundaries by advection and dispersion.

Stationary coffee skeleton

$$dM_a = m_a A dz, \quad dM_d = m_d A dz$$

Release and capture exchange fines locally between water and skeleton.

The highlighted slice defines the continuum balance. It is not a numerical cell.

Presenter transcript

Consider an infinitesimally thin slice through the puck with cross-sectional area A and thickness dz . Within this volume, the water-filled pores contain mobile fines, while the stationary skeleton carries retained fines. Advection and dispersion act across the two axial boundaries. Release, capture and remobilization exchange fines locally between the two phases. Taking the slice thickness to zero gives the continuum equations. Finite-volume cells are a later numerical choice. Extraction chemistry is outside this model; its hydraulic relevance remains an assumption to assess.

Continuum equations and optional surface loading



$$\frac{\partial(\varepsilon c)}{\partial t} + \frac{\partial(qc)}{\partial z} = \frac{\partial}{\partial z} \left(\varepsilon D \frac{\partial c}{\partial z} \right) + R_{\text{rel}} - R_{\text{dep}} + R_{\text{rem}},$$
$$\frac{\partial m_a}{\partial t} = -R_{\text{rel}}, \quad \frac{\partial m_d}{\partial t} = R_{\text{dep}} - R_{\text{rem}}.$$

Equivalent surface description

$$m_a = \sigma c_a, \quad m_d = \sigma c_d, \quad [\sigma] = \text{m}^2/\text{m}_{\text{bulk}}^3, \quad [c_a] = [c_d] = \text{kg}/\text{m}^2.$$

Use $\partial_t(\sigma c_a)$ if the specific surface area changes.

Only mobile fines travel axially. All R terms have units $\text{kg}/(\text{m}_{\text{bulk}}^3 \text{ s})$.

Presenter transcript

These are the balance equations before choosing a grid or numerical method. The mobile phase gains fines from initial release and deposit remobilization, and loses fines by capture. Heiko's surface-loading formulation is equivalent: define sigma as accessible surface area per bulk volume and c_a as mass per unit surface. Then m_a equals sigma times c_a. If sigma changes, its derivative must remain inside the time derivative. The existing solver stores bulk mass densities; introducing measured surface area would clarify parameter interpretation without requiring a new conservation law.

Release and capture are constitutive hypotheses

Release of initially attached fines

$$R_{\text{rel}} = k_{\text{rel}} m_a (|q|/q_{\text{ref}})^{a_r}, \quad [k_{\text{rel}}] = \text{s}^{-1}.$$

Capture of mobile fines

$$R_{\text{dep}} = k_{\text{dep}} \chi(z) |q| c \left[1 - \frac{m_d}{m_{\text{cap}}} \right]_+, \quad [k_{\text{dep}}] = \text{m}^{-1}.$$

- Release depends on the remaining available inventory and flow.
- Capture depends on mobile flux and available deposit capacity.
- $\chi = 1$ is the unbiased baseline. Axial weighting needs evidence.
- Compare these laws using multiple flow rates and spatial deposit data.



Presenter transcript

These equations are closures, not consequences of mass conservation. The release law assumes that stronger flow increases the release rate and that the remaining available inventory limits release. The capture coefficient has units of inverse length because it multiplies a mass flux to give a volumetric rate. The capacity factor reduces capture as deposits accumulate. The optional axial weighting should not impose a basket-side blocking region without supporting evidence. We can replace these laws without changing the conservation framework.



Deposit remobilization and local pore accessibility

Deposits return to the water only above a forcing threshold

$$R_{\text{det}}^* = k_{\text{det}} m_d \left[\frac{G}{G_{\text{crit}}} - 1 \right]_+^b, \quad G = \left| \frac{\partial p}{\partial z} \right|.$$

At $G \leq G_{\text{crit}}$, remobilization is zero. Above the threshold, more deposits and stronger forcing increase the potential detachment rate.

$$R_{\text{rem}} = P_{\text{mob}} R_{\text{det}}^*, \quad P_{\text{mob}} = \frac{1}{1 + \exp[s(d_f/d_h - \lambda_c)]}.$$

- P_{mob} : fraction that can enter flowing pores locally.
- Larger fines d_f relative to pore diameter d_h reduce this fraction.
- λ_c : size ratio at 50% access. s : transition steepness.
- The remaining fraction stays deposited. Mobile fines may deposit again.

Notation: $R_{\text{rem}} = R_{\text{wash}}$ and $P_{\text{mob}} = P_{\text{esc}}$ in the code. The logistic law is an unvalidated closure.

Presenter transcript

Remobilization requires deposits and a pressure-gradient magnitude above G_{crit} . The positive-part operation makes the potential rate zero below that threshold. Above it, k_{det} sets the rate scale and b controls sensitivity to excess forcing. P_{mob} is the effective fraction that can enter flowing pores locally. It falls when fine diameter becomes large relative to the hydraulic pore diameter. λ_c is the ratio giving fifty-percent accessibility, and s controls steepness. The remaining fraction stays in the deposit inventory. Remobilized fines can be captured again downstream. We use clearer slide symbols R_{rem} and P_{mob} for the existing code quantities R_{wash} and P_{esc} ; this is a notation change only. The logistic law remains unvalidated.



Fines leaving the puck: sieve retention and cup collection

The outlet carries mobile fines out of the puck

$$\dot{M}_{\text{out}}(t) = Q(t)c(L, t) \quad [\text{kg/s}].$$

$c(L, t)$ is the fines concentration at the puck outlet. The outlet model neglects dispersive exchange with the basket.

Retained by the sieve

$$\dot{M}_{\text{sieve}} = \eta_s \dot{M}_{\text{out}}$$

The basket retains the fraction η_s of the outgoing fines.

$$M_j(t) = \int_0^t \dot{M}_j(\tau) \, d\tau, \quad j \in \{\text{out}, \text{sieve}, \text{cup}\}, \quad M_{\text{out}} = M_{\text{sieve}} + M_{\text{cup}}.$$

Zero initial collected masses. The illustrative runs use constant $\eta_s = 0.35$, so 65% reaches the cup.

Retention η_s requires calibration. It differs from hydraulic resistance R_s . In the current model, sieve loading does not change R_s .

Collected in the cup

$$\dot{M}_{\text{cup}} = (1 - \eta_s) \dot{M}_{\text{out}}$$

The remaining fraction passes through the basket into the cup.

Presenter transcript

After remobilization we follow fines to the final boundary. The outgoing mass rate equals volumetric water flow Q times the mobile-fines concentration in the last cell, representing c at the puck outlet. Q has units cubic metres per second and c kilograms per cubic metre of water, so their product is kilograms per second. No extra porosity factor is needed because Q is already volumetric flow, equal to area times superficial velocity. The open-outlet closure omits dispersive exchange. The sieve retains η_s times this rate, and one minus η_s enters the cup. Integrating each rate from zero gives cumulative mass. Thus outlet mass equals sieve-retained mass plus cup mass. We assume constant retention of 35 percent in these illustrative runs; this requires experimental calibration. Retention is distinct from hydraulic resistance R_s . We do not yet increase basket resistance as sieve fines accumulate. Cup fines are only part of what left the puck, and leaving the puck is only part of what was mobilized internally.

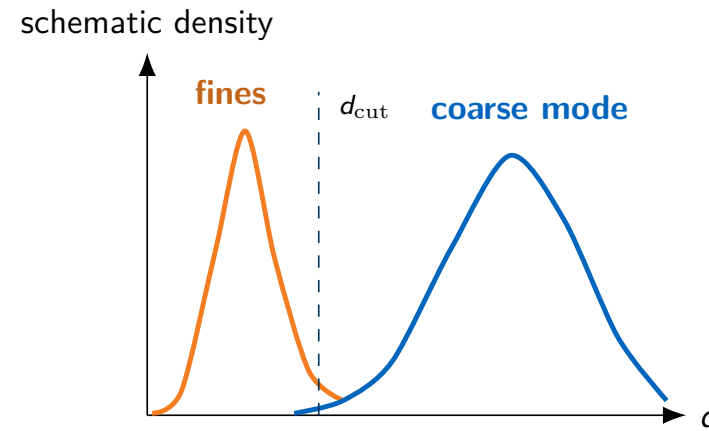


The PSD is reduced to quantities used by the 1D closures

Measured bimodal PSD

- Fit a five-parameter mixture model.
- Choose a mobility cutoff d_{cut} .
- Estimate the fine fraction and its releasable share.
- Calculate representative fine and coarse diameters.

Use in the model: the fine mass fraction and its releasable share set the initial attached inventory $m_{a,0}$. The fine diameter d_f controls local pore access. A representative bed diameter helps estimate the initial pore size.



Source: Bora and Briesen, Journal of Food Process Engineering 49 (2026), e70645.

Presenter transcript

The particle-size distribution provides an estimate of the fine fraction and representative diameters. A diameter cutoff alone does not prove that every particle below it is releasable. An availability fraction or independent release measurement is needed to initialize m_a . The two curves are schematic. Specific surface area should preferably be measured or estimated consistently from the distribution and particle shape. A pore-network calculation could later help connect particle size to accessibility, but it is not a current data source. The bottom text explains the parameter mapping in words: fine mass and releasable share initialize the attached inventory, fine diameter enters local accessibility, and a representative bed diameter informs initial pore size.



Constant pressure or PI control

Constant pressure: controlled reference

$$\Delta p_{\text{total}} = 9 \text{ bar}, \quad Q = \Delta p_{\text{total}} / R_{\text{tot}}(t).$$

The source supplies whatever flow maintains 9 bar. This represents an experimental 9 bar plateau only when the machine can sustain that pressure.

Flow-limited PI: machine–puck prediction

$$e = p_{\text{set}} - \Delta p_{\text{total}}, \quad Q_{\text{command}} \leq Q_{\text{max}}.$$

The controller adjusts pump flow, with feedforward and pump lag. A permeable puck may stay below the target because the pump reaches its flow limit.

PI represents machine constraints. Its predictive value requires representative pump limits and controller settings.

Presenter transcript

Constant pressure and PI answer different questions. Constant nine bar is a controlled reference that assumes enough water supply to maintain that pressure, even for a coarse puck. Flow-limited PI includes finite pump delivery and actuator lag, so it can represent a machine that fails to attain the target. This makes PI more representative than an unlimited pressure source when predicting machine-puck behavior, provided the pump limits and control settings match the machine. The current comparison uses illustrative settings and is not machine validation. The solver default is a software choice, not a scientific preference.



Pressure reference and experimental comparison

Our priority: validate flow and cup fines using measured pressure

- Prescribe the actual pressure drop to test the puck model without fitting uncertain machine-control parameters.
- Constant 9 bar is the reference for a sustained 9 bar interval. Use the actual pressure for experiments below the target.
- Retain flow-limited PI to predict pressure and flow for different initial PSDs, with a representative machine model.

$p_{\text{out}} = 0$, $p_{\text{puck exit}} = QR_s$. The measured pressure must refer to the same endpoints as the model.

Measured pressure for validation. PI for machine–puck prediction. Constant 9 bar for comparison.

Measured time-series input remains an extension. The current saturated-puck model excludes initial wetting.

Presenter transcript

For our main validation goal we should supply measured pressure and predict outlet flow and cup fines. This avoids compensating for an incorrect machine model by fitting fines kinetics. Constant pressure is appropriate for a measured plateau, at nine bar or the actual lower pressure. A measured time-series input remains to be implemented. PI supports prediction across initial particle-size distributions without supplying the experimental pressure, but still needs representative machine parameters and calibrated puck closures. All pressures must use consistent sensor endpoints: R_s is the basket resistance and Q times R_s is the pressure just above it relative to the cup. Starting only at the plateau requires a fines state consistent with the preceding ramp. Initial wetting remains outside the saturated model.



Boundary conditions for clean feed and an open outlet

$$J = qc - \varepsilon D \partial_z c$$

Inlet: no fines supplied

$$J(0, t) = qc_{\text{in}} = 0$$

The feed is clean water. We prescribe the total incoming fines flux.

This does not require imposing both $c = 0$ and $\partial_z c = 0$ on the interior solution.

In the code, boundary dispersive fluxes are omitted; inlet total flux is zero and outlet flux is qc . Reconsider this closure if downstream mixing or back-dispersion matters.

Outlet: convective removal

$$\partial_z c(L, t) = 0, \quad J(L, t) = qc(L, t)$$

We neglect dispersive exchange with the downstream basket/cup region.

This open-outlet approximation allows fines to leave by advection, without a modeled downstream concentration field.

Presenter transcript

The clean-water inlet supplies no fines, so we specify zero total incoming fines flux. This is a flux boundary condition; it does not mean imposing both zero concentration and zero concentration gradient on the interior solution. At the outlet, we neglect dispersive exchange with the downstream region and let advection carry fines out. The code omits the dispersive contribution at boundary faces. These choices close the truncated puck domain without modeling a downstream concentration field. They are assumptions, not a requirement of finite volumes. If downstream mixing or back-dispersion affects the result, the boundary model must be revisited. Initial fines fields remain necessary, and basket resistance sets the outlet pressure relative to the cup.



Section 4 of 7

Numerical solution and verification

Algebraic Darcy hydraulics and finite-volume fines transport

Presenter transcript

Section 4 of 7: Numerical solution and verification.
Algebraic Darcy hydraulics and finite-volume fines transport.



Darcy flow follows from the current hydraulic resistance

At each update, use the current cell permeabilities $K_i(t)$

$$R_{\text{puck}} = \frac{1}{A} \int_0^L \frac{\mu}{K(z, t)} dz \approx \frac{\mu}{A} \sum_i \frac{\Delta z_i}{K_i}, \quad R_{\text{tot}} = R_s + R_{\text{puck}}.$$

Operating mode	Algebraic hydraulic evaluation
Pressure imposed	$Q = \Delta p_{\text{total}} / R_{\text{tot}}$
Flow imposed	$\Delta p_{\text{total}} = QR_{\text{tot}}$
PI control	Pump dynamics supply $Q(t)$, then $\Delta p_{\text{total}} = Q(t)R_{\text{tot}}$.

R has units Pa s/m^3 . The cell sum evaluates the resistance integral.

Darcy hydraulics use algebraic relations. FVM advances the fines-transport balance.

Presenter transcript

At each time step, the deposited fines determine the current permeability. We sum the Darcy resistance of the axial cells and add the basket resistance. With imposed pressure, flow is pressure divided by total resistance. With imposed flow, pressure is flow times resistance. In PI mode, the pump state evolves in time and supplies the actual flow. This is an algebraic hydraulic calculation, whereas finite volumes advance the fines balance. The resistance integral is evaluated numerically as a cell sum, so the calculation still uses the mesh. We do not solve a separate momentum PDE.



Velocity and pressure profiles follow from Darcy

Velocity from the common flow rate

$$q(t) = \frac{Q(t)}{A}, \quad u_{\text{pore},i}(t) = \frac{q(t)}{\varepsilon_i(t)}.$$

q is the superficial velocity, uniform along the constant-area puck.

The local pore velocity changes with porosity.

Pressures are relative to the cup outlet. The recurrence gives face pressures, with cell pressure at the midpoint for a cellwise constant gradient.

Pressure from the local resistance

$$G_i = \frac{\mu q}{K_i}, \quad p(L) = QR_s.$$

$$p_{i-1/2} = p_{i+1/2} + G_i \Delta z_i.$$

Starting at the basket, accumulate each cell's pressure loss upstream.

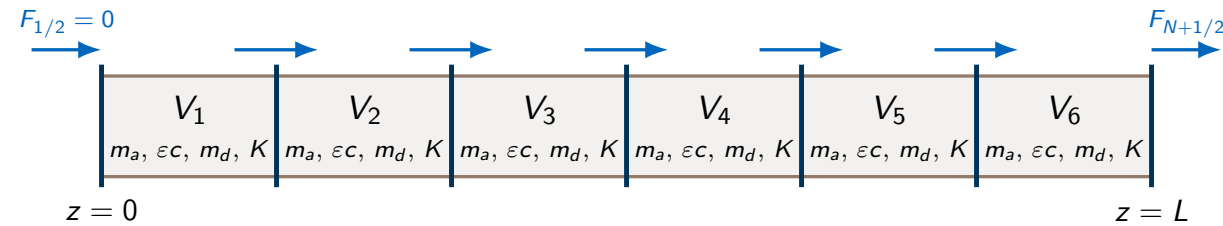
Numerical quadrature reconstructs pressure. No separate momentum or pressure PDE is solved.

Presenter transcript

Once the common volumetric flow is known, superficial velocity is Q divided by puck area. The mean pore velocity in each cell is superficial velocity divided by local porosity. For pressure, begin at the downstream face, where basket resistance gives Q times R_s relative to the cup. Accumulate μ times q divided by K times cell thickness upstream. This numerical quadrature reconstructs face pressures directly. There is no additional pressure or momentum PDE, and no transient fluid momentum state. Fines and pump dynamics can nevertheless evolve with time.



The finite-volume method enforces flux balance cell by cell



$$\frac{(\varepsilon c)_i^{n+1} - (\varepsilon c)_i^n}{\Delta t} = - \frac{F_{i+1/2} - F_{i-1/2}}{\Delta z} + R_{\text{rel},i} - R_{\text{dep},i} + R_{\text{rem},i}.$$

$$F_{i+1/2} = \underbrace{q c_i}_{\text{advection}} - \underbrace{\varepsilon_{i+1/2} D \frac{c_{i+1} - c_i}{\Delta z}}_{\text{dispersion}}, \quad q \geq 0.$$

First-order upwind advection and centred dispersion. No high-order flux limiter.

Shared face fluxes cancel between adjacent cells, preserving the global mass balance.

Presenter transcript

Finite volumes balance the fines inventory in each cell through shared face fluxes. The flux that leaves one cell enters its neighbor, so interior exchanges cancel in the global sum. The code uses first-order upwind advection and centered dispersion. It does not need a high-order advection flux limiter, but numerical diffusion can smear steep concentration fronts. We therefore assess spatial refinement. Positivity-preserving local reaction transfers are a separate issue from transport flux limiting. The next slide explains how the timestep is selected.

Why finite volumes for fines transport?



The main requirement is consistent mass accounting

- Each cell stores a fines inventory and exchanges mass through its faces.
- One shared face flux leaves one cell and enters the next.
- Summing the cells cancels internal fluxes; boundary fluxes give collected fines.
- Local release, capture and remobilization transfer mass between inventories.

Alternative	Assessment for this model
Finite differences	Equally viable in conservative flux form; on a uniform grid, they can give the same update.

FVM is a transparent first implementation. Its first-order upwind diffusion requires grid refinement.

Presenter transcript

We chose finite volumes because shared face fluxes make local and global fines mass accounting explicit. Mass leaving one cell enters its neighbor, and the outlet flux determines collected fines. Conservative finite differences can be equally appropriate and can yield the same update on a uniform grid. FVM is therefore a transparent implementation choice rather than the only valid method. First-order upwind transport adds numerical diffusion, so grid refinement is still necessary.

The smallest timescale limits the numerical step

$$\Delta t = \min(\Delta t_{\max}, \Delta t_{\text{transport}}, \Delta t_{\text{pump}}, t_{\text{end}} - t)$$

$$\Delta t_{\text{transport}} = \frac{0.42}{|q|/(\varepsilon_{\min}\Delta z) + 2D/\Delta z^2}$$

Limit	Purpose
$\Delta t_{\max}$	User-selected upper bound for temporal resolution.
$\Delta t_{\text{transport}}$	Combined explicit advection–dispersion restriction. Here $\varepsilon_{\min} = \min_i \varepsilon_i(t)$.
$\Delta t_{\text{pump}} = \tau_p/4$	Used only in PI mode to resolve pump response; this is not the PI integral time.
$t_{\text{end}} - t$	Stops exactly at the requested final time.

For $D = 0$: $\text{Co} = |q|\Delta t/(\varepsilon_{\min}\Delta z) \leq 0.42$. Timestep refinement still checks accuracy of kinetics and coupling. The transport bound is the implemented estimate, not a general heterogeneous-field proof.

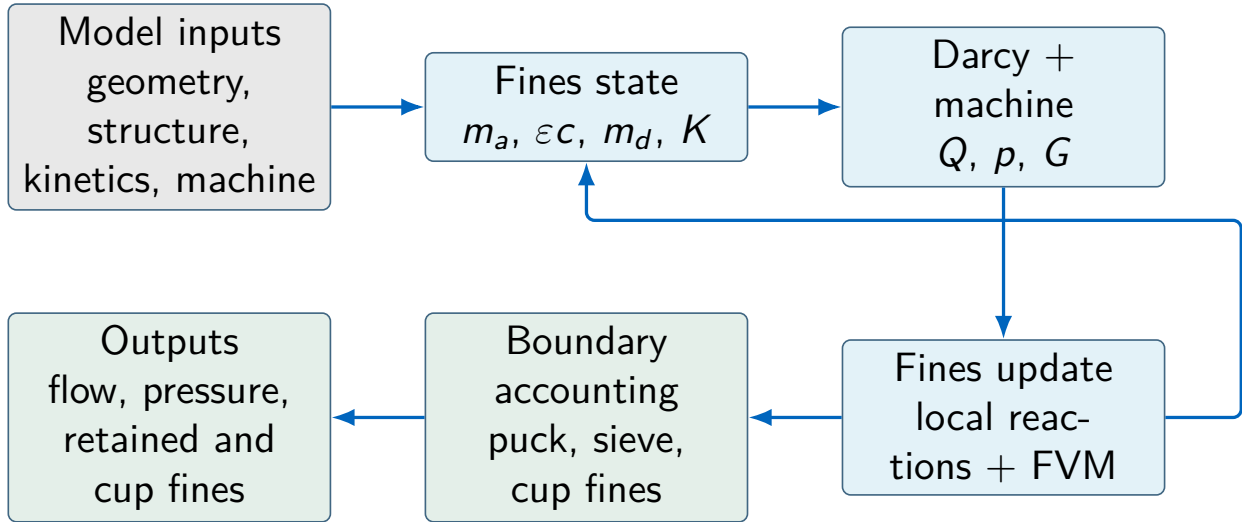


Presenter transcript

The solver takes the smallest of four limits. The user maximum controls the largest permitted step. The transport estimate combines advection at the largest pore velocity with explicit dispersion and uses a safety factor of 0.42. In PI mode, another bound takes one quarter of the pump response time. This resolves the actuator response and is not a bound based on the PI integral time. The remaining simulation time prevents overshooting the endpoint. Exponential local transfers preserve nonnegative inventories, but this does not guarantee temporal accuracy of their coupling. Timestep refinement remains necessary, and strong porosity variation warrants checking the transport estimate against the actual face coefficients.



Coupled physics and its numerical update



Physical coupling: deposit changes ϵ , K and d_h ; these change pressure and flow; the new hydraulics then alter release, deposition and washout.

Quasi-steady flow is a physical assumption. Update ordering is a numerical choice.

Presenter transcript

The feedback is physical: deposits change resistance, resistance changes flow and pressure gradients, and these alter fines transport and transfer. The equations define this coupling regardless of how we solve them. The code evaluates dependencies in a sequence during each timestep. That numerical ordering introduces discretization error and should be checked by timestep refinement.



Numerical verification and physical validation

Verification

- Conservation and nonnegative inventories.
- Analytical hydraulic limiting cases.
- Grid and timestep convergence.
- Controller limits and boundary accounting.

Validation

- Pressure and flow measurements.
- CT structural changes.
- Collected fines and size distributions.
- Predictions for withheld cases.

Current unit tests: 17/17 passed. The four pressure-mode runs conserve fines mass to relative error below 2.1×10^{-15} . Physical validation remains outstanding.

Presenter transcript

All seventeen unit tests pass. The four matched pressure-mode runs conserve fines mass with relative error below 2.1 times ten to the minus fifteen. Tests verify exact constant pressure across a nonuniform puck and during deposition. PI tests verify target tracking and persistent pump saturation for a coarse puck without evolving fines. These are numerical checks, not experimental validation.



Section 5 of 7

Illustrative results and model sensitivity

How strongly do the hydraulic and permeability choices matter?

Presenter transcript

Section 5 of 7: Illustrative results and model sensitivity.
How strongly do the hydraulic and permeability choices matter?.

Darcy gives practically the same results

We ran the same case twice, changing only the flow law.

Result at 60 s	Darcy–Forchheimer	Darcy
Pressure [bar]	8.9906	8.9906
Flow rate [mL/s]	0.04535	0.04535
Cup fines [g]	0.03993	0.03993

Values rounded for display.

Across the entire simulation, the largest pressure difference was only **0.0203 Pa**.

We therefore use Darcy for this tested case.

Controlled 60 s simulations with identical exponential permeability, kinetics and PI settings.



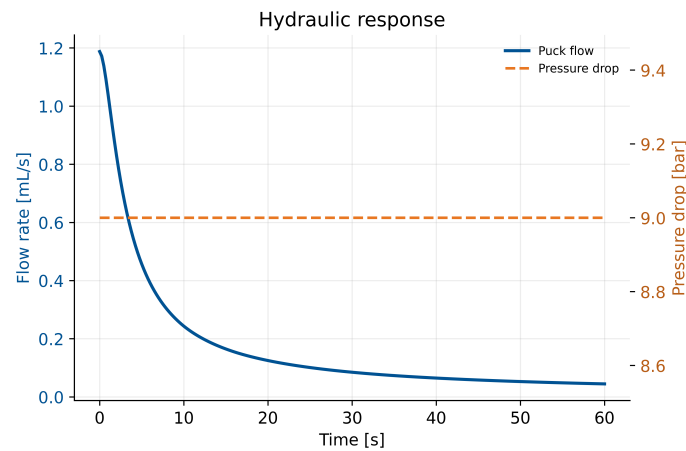
Presenter transcript

We ran the same case twice, changing only the flow law. One run includes the Forchheimer inertial term and the other uses Darcy alone. The permeability closure, initial state, kinetics and PI controller settings are identical. At sixty seconds, pressure, flow rate and collected cup fines are the same to the precision displayed in the table. They are not mathematically identical: the values have been rounded. Across the entire simulation, the largest pressure difference is only 0.0203 pascals. We therefore choose Darcy for this tested case because including inertia makes no meaningful difference to these predictions.



The illustrative case develops strong viscous blockage

Constant 9 bar across puck + basket. Exponential permeability.



At 60 s

- total pressure: 9.00 bar;
- flow: 0.0447 mL/s;
- minimum K/K_0 : 0.0125.

Interpretation

At fixed 9 bar, deposition increases resistance and reduces flow. No PI controller is needed.

New Darcy baseline: exponential permeability, constant 9 bar. Illustrative parameters, no experimental calibration.

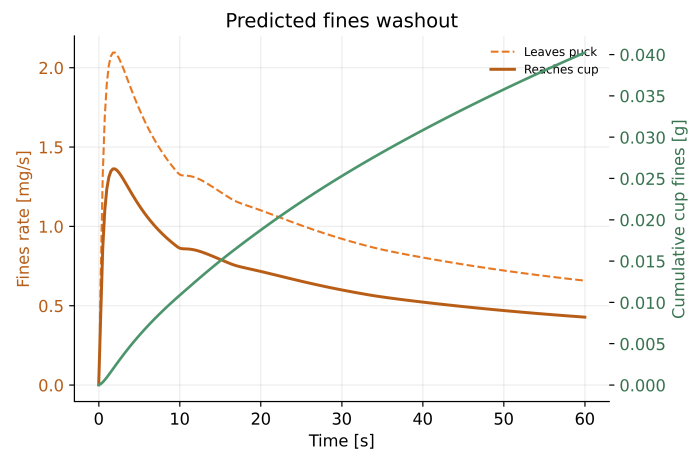
Presenter transcript

This is the new constant-pressure baseline with Darcy and exponential permeability. Total pressure remains exactly nine bar. At sixty seconds the flow is 0.0447 millilitres per second and the minimum local permeability ratio is about 0.0125. Deposition increases resistance, so flow falls even with no PI controller. The strong loss depends on the assumed exponential closure.



The model distinguishes puck washout from fines in the cup

Constant 9 bar across puck + basket. Exponential permeability.



Cumulative masses

- puck outlet: 0.0619 g;
- basket retained: 0.0217 g;
- cup: 0.0402 g.

With $\eta_s = 0.35$, the basket retains 35% of the fines that leave the puck.

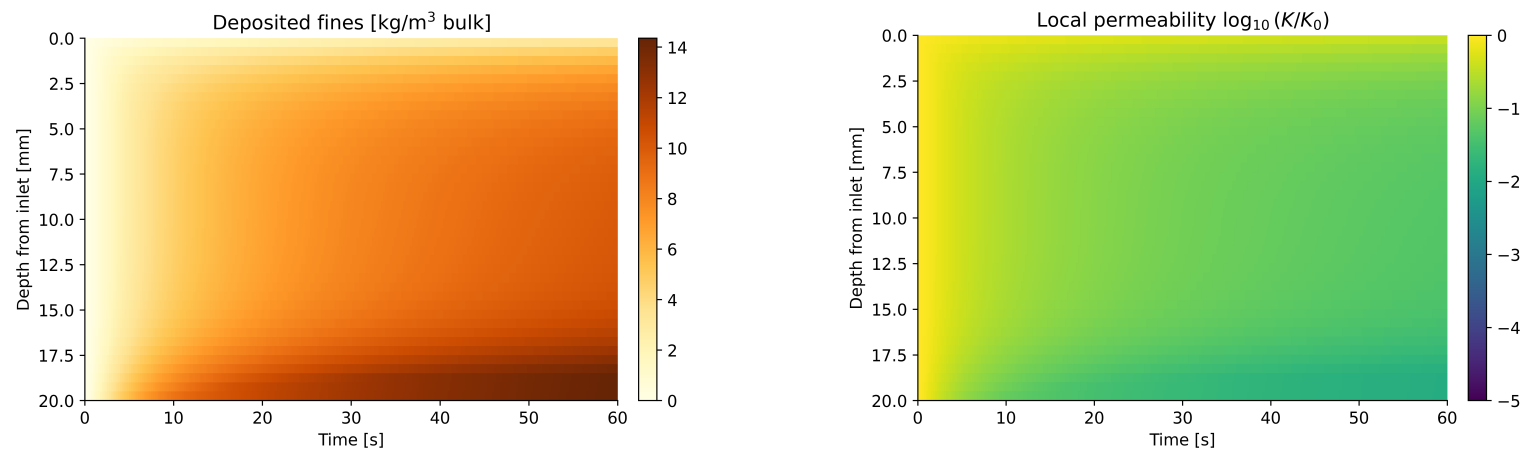
Constant 9 bar Darcy baseline with exponential permeability; constant basket retention.

Presenter transcript

The new constant-pressure baseline gives 0.0619 grams leaving the puck, 0.0217 grams retained by the basket, and 0.0402 grams reaching the cup. The fixed basket retention fraction is 35 percent. These are illustrative simulated quantities, not measurements.



The axial fields locate the effective blocking region
Constant 9 bar across puck + basket. Exponential permeability.



These fields can be compared with interrupted CT reconstructions, but a 1D layer cannot represent lateral channel geometry.

Constant 9 bar Darcy baseline with exponential permeability. The later closure comparison retains its original PI settings.

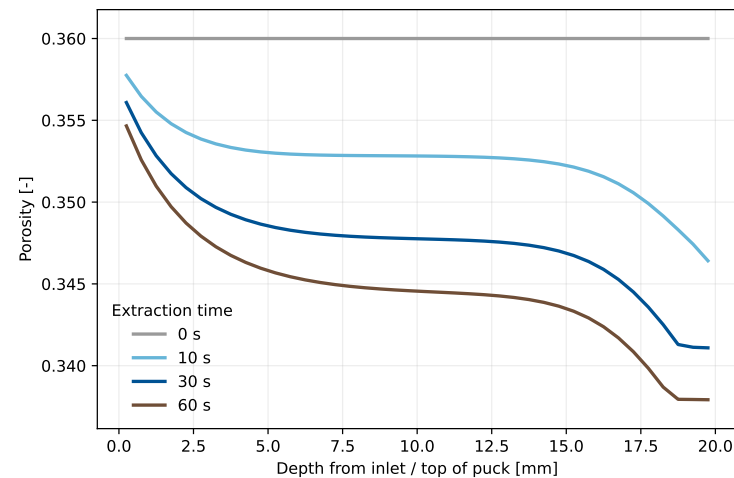
Presenter transcript

These fields now come from the constant nine-bar Darcy baseline with exponential permeability. They show effective axial deposition and permeability change. They do not resolve radial channels. The subsequent closure sensitivity study is the archived PI comparison, which is labeled separately. The visible label identifies constant nine bar across puck plus basket and exponential permeability. These are coupled fields from that single run, not universal deposit or permeability profiles.



Porosity profile evolves as fines redistribute

Constant 9 bar across puck + basket. Exponential permeability. Illustrative, uncalibrated case.



Model $t = 0$: start of saturated extraction; dry/preinfusion not modeled.

Predicted structure

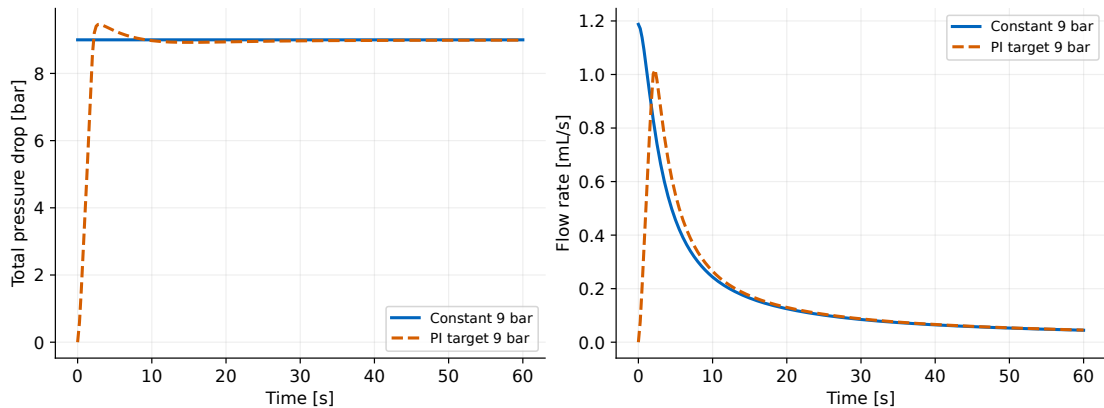
- Direct model observable: $\varepsilon(z, t)$.
- Initial profile: $\varepsilon_0 = 0.36$.
- Deposition lowers porosity, most strongly near the outlet/bottom.
- At 60 s: $\varepsilon \approx 0.338\text{--}0.355$.

The predicted axial porosity evolution can be compared directly with interrupted CT profiles at matched extraction times.

Presenter transcript

This plot uses the axial porosity history already returned by the solver. The initial illustrative bed is uniform at epsilon 0.36. As fines are released, transported and recaptured, deposition lowers porosity throughout the puck, with the strongest decrease near the outlet because the current capture closure is weighted there. At sixty seconds the profile spans about 0.338 to 0.355. This is not a dry-to-wet comparison: model time zero is the start of full extraction in an already saturated one-dimensional bed. The direct validation target is therefore the CT porosity profile at matched extraction times.

Fine puck: constant pressure versus PI
Constant 9 bar vs. PI, 9 bar target. Exponential permeability.



At 60 s: constant pressure gives $Q = 0.0447$ mL/s; PI gives $Q = 0.0454$ mL/s. The PI startup ramp changes the earlier fines history.

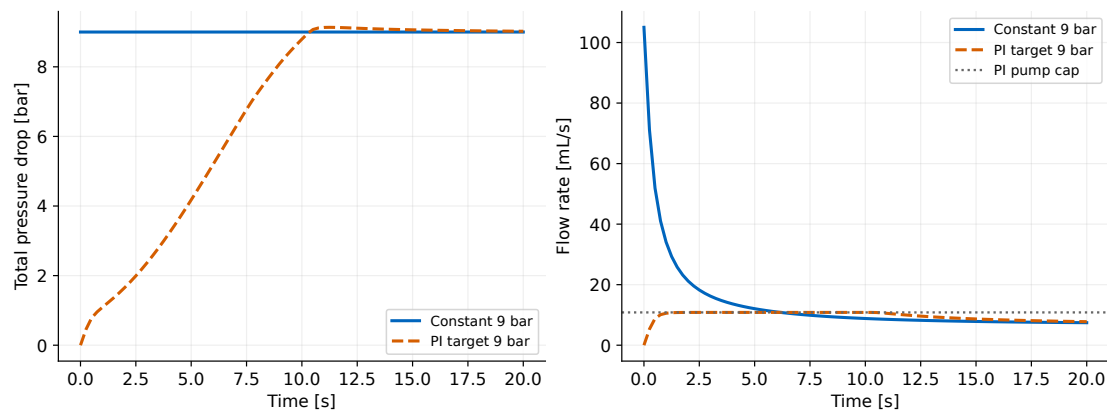
Matched 60 s runs. Only control mode changes. Darcy + exponential, illustrative parameters.

Presenter transcript

These runs use identical physical parameters and differ only in operating mode. Constant pressure starts at nine bar immediately. PI ramps its target and adjusts pump flow with lag. Their final flow rates are close, but early flow and fines histories differ. The PI settings are illustrative, not a calibrated representation of the experimental machine.

Coarse puck: the pump cap limits initial pressure

Constant 9 bar vs. PI, 9 bar target. Exponential permeability.



Constant 9 bar initially requires 105 mL/s, above the PI cap of 10.83 mL/s. PI reaches only 1.66 bar at 2 s. Deposition later raises resistance enough to reach the target.

Matched 20 s runs with active fines. This coarse case does not stay below 9 bar throughout. No experimental calibration.



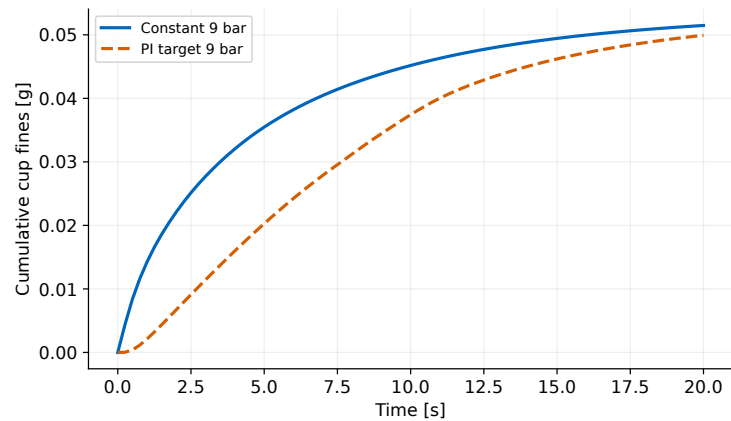
Presenter transcript

The highly permeable initial coarse puck would require about 105 millilitres per second at nine bar. Constant pressure supplies that mathematical flow without a cap. PI is limited to 10.83 millilitres per second, giving only 1.66 bar at two seconds. Fines then deposit, resistance rises, and the target becomes attainable. This illustrates temporary saturation, not an experiment that remains below nine bar for the whole extraction.



Pressure history also changes fines reaching the cup

Constant 9 bar vs. PI, 9 bar target. Exponential permeability.



Same coarse puck and kinetics

Different pressure histories change flow, release and deposition.

A fixed 9 bar run cannot directly represent a lower-pressure experiment.

Only control mode differs. Basket retention and permeability closure are identical.

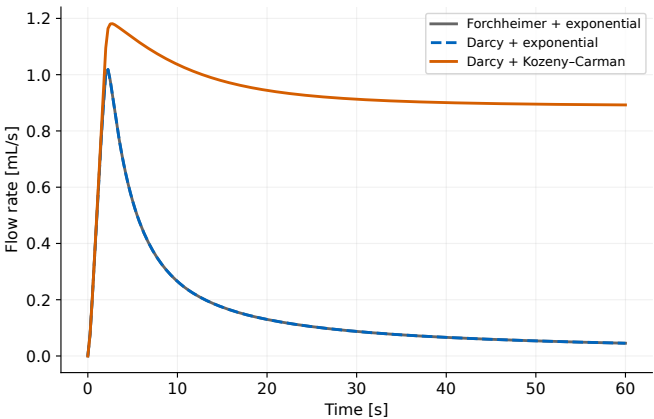
Presenter transcript

Changing only the operating mode changes the entire hydraulic history. That affects release, capture and transport, and therefore cumulative fines in the cup. We must match the experimental pressure conditions before adjusting kinetic parameters. Otherwise we might compensate for an incorrect boundary condition by fitting the fines model.



Permeability closure strongly changes the flow response

Flow-limited PI, 9 bar target. Exponential vs. Kozeny–Carman.



At 60 s

Exponential:
 $Q = 0.0454 \text{ mL/s}$

Kozeny–Carman:
 $Q = 0.892 \text{ mL/s}$

Both Darcy runs approach
the 9 bar PI target.

The closure changes resistance and flow, even with nearly the same final pressure.

Same initial K_0 , kinetics and machine parameters. Gray and blue curves overlap: inertia is negligible.

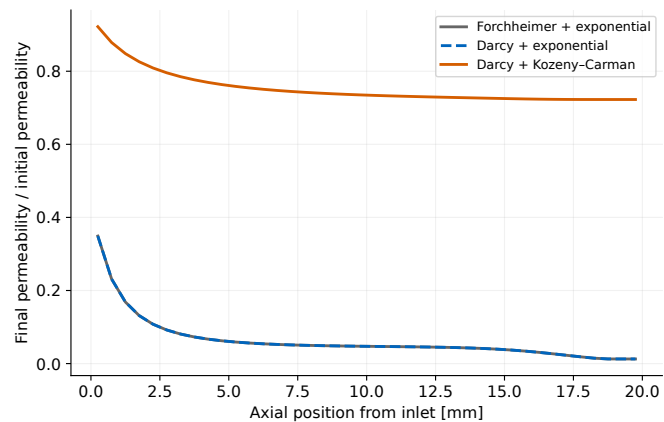
Presenter transcript

The blue Darcy-exponential curve overlaps the gray Forchheimer reference. The orange curve changes only the permeability closure to Kozeny-Carman. At sixty seconds the flow is about 0.892 milliliters per second, compared with 0.0454 for exponential permeability. Both PI-controlled cases end near nine bar, so pressure alone hides a substantial difference in resistance and flow. This illustrates why simultaneous pressure and flow measurements are needed. All rates and controller settings were kept unchanged.



Kozeny–Carman predicts weaker permeability loss

Flow-limited PI, 9 bar target. Exponential vs. Kozeny–Carman.



Minimum K/K_0

Exponential:
0.0126

Kozeny–Carman:
0.723

The same deposit-to-porosity
law feeds both closures.

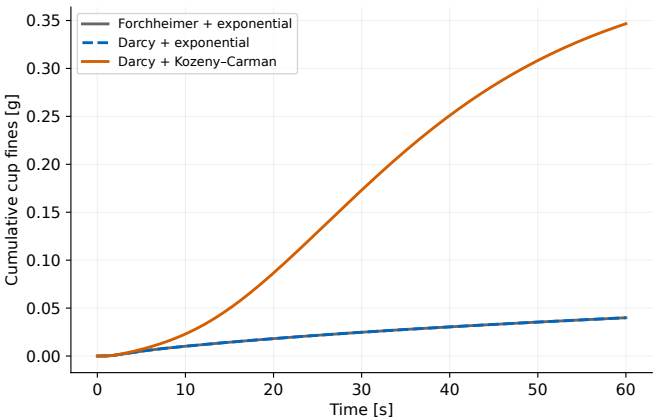
The exponential law imposes stronger blockage. The comparison does not establish which closure is correct.

Final profiles at 60 s. Coupled deposit fields also differ because flow feeds back on kinetics.

Presenter transcript

The exponential law produces a minimum relative permeability of about 0.0126. The implemented Kozeny-Carman relation gives about 0.723. The latter therefore predicts much weaker permeability loss for this material-state closure. These are coupled runs: different flow also produces different deposit profiles. The result identifies strong sensitivity to the structural closure, but does not establish which law is physically right. We must compare with measured hydraulic and structural changes rather than tune one closure to match the other. The visible label identifies flow-limited PI with a nine-bar target and the two permeability closures. Different coupled deposition histories mean these profiles do not represent applying both laws to one fixed deposit field.

The permeability closure also changes cup fines
Flow-limited PI, 9 bar target. Exponential vs. Kozeny–Carman.



Cup fines at 60 s

Exponential:
0.0399 g

Kozeny–Carman:
0.3466 g

Same basket retention:
 $\eta_s = 0.35$.

This is closure sensitivity, not validation. Neither prediction is an experimental target.

All kinetic and machine parameters held fixed. No fitting to reproduce the exponential-law result.

Presenter transcript

Cumulative cup fines increase from about 0.0399 grams for exponential permeability to about 0.3466 grams for Kozeny-Carman. The basket retention fraction remains 35 percent in both. Higher sustained flow changes release and transport, which changes the collected mass even though the kinetic coefficients are identical. This is a sensitivity result using illustrative inputs. Neither simulated curve is a substitute for experimental calibration and withheld-case validation.



Section 6 of 7

Calibration and validation strategy

Which measurements constrain each part of the model?

Presenter transcript

Section 6 of 7: Calibration and validation strategy.
Which measurements constrain each part of the model?.

Calibration begins with independently measured structure



Input	Measurement or proposed inference
Geometry and fluid	Puck height and area, temperature-dependent viscosity and density.
PSD and available fines	Fine fraction, representative sizes and an independently constrained releasable share.
Initial structure	CT porosity and heterogeneity. Accessible specific surface σ with resolution uncertainty.
Hydraulics	Low-flow pressure–flow slope for K_0 , empty-basket loss for R_s .
Future pore net-works	Connectivity-sensitive permeability and capture/accessibility closures.

CT-derived hydraulic diameter is a structural proxy, not a direct measurement of every pore throat.

Presenter transcript

We should determine as much as possible independently before fitting the kinetics. Geometry and fluid properties are straightforward inputs. PSD informs fine mass and representative sizes, but the releasable fraction needs separate attention. CT gives porosity and spatial variation, while surface area is sensitive to image resolution and segmentation. Low-flow pressure-flow data constrain permeability. Pore-network calculations could later provide connectivity-sensitive closures. We should not present those future results as available measurements.



Experiments to distinguish competing kinetic models

- Fines collected in time intervals at several flows constrain release and capture.
- Interrupted CT locates retained material and redistribution.
- Pressure pulses test deposit remobilization and threshold behavior.
- Outlet size distributions test size selectivity.
- Basket-retained mass separates puck transport from cup collection.

Model comparisons One versus two retained populations; accessibility factor versus $P_{\text{mob}} = 1$; uniform capture versus axial weighting.

These are proposed experiments and comparisons. Validate on withheld pucks or grind settings.

Presenter transcript

The experiments should distinguish mechanisms rather than merely provide more fitting points. Time-resolved fines at several flow rates help separate release from capture. Interrupted CT provides spatial evidence that cup mass alone cannot supply. Pressure pulses probe remobilization. We should explicitly compare a simpler one-inventory model with the two-inventory model, and test whether the accessibility factor is needed. These are proposed measurements. Their availability has not been established, and the final evaluation should use withheld cases.



Next comparison: redistribution versus washout

Retained redistribution Fines release and move, but deposition elsewhere in the puck dominates. CT changes while cumulative outlet mass stays small.

Net washout Mobile fines reach the puck outlet. Basket retention and cup collection quantify the outgoing mass.

Proposed study

- Compare pressure or flow levels with the same initial puck structure.
- Record M_a , M_m , M_d and M_{out} with axial deposit profiles.
- Darcy is sufficient here. Compare both permeability predictions with data.
- Use PNM or 2D/3D only where pore connectivity or radial bypass demands it.

Presenter transcript

Heiko suggests a particularly useful target: conditions where fines move internally but remain trapped, compared with conditions where they leave the puck. Low cup mass alone cannot distinguish no release from release followed by recapture. We need both the inventory history and spatial deposit changes. The closure study is complete. We should next compare operating levels and test the competing closure predictions against measurements. More detailed pore networks or multidimensional models should answer a demonstrated limitation rather than be assumed prerequisites.



Section 7 of 7

Conclusions and next steps

What the model shows and which evidence is still needed

Presenter transcript

Section 7 of 7: Conclusions and next steps. What the model shows and which evidence is still needed.



Conclusions and next steps

- Fines inventories evolve through conservative transport and local exchange.
- Darcy gives pressure and flow from the evolving hydraulic resistance.
- Inertia is negligible in the tested case, while permeability closure strongly affects the predictions.
- Calibration needs independent structure, pressure–flow and fines measurements.

The next step is to constrain the permeability and kinetic closures with data.
Physical validation remains outstanding.

Presenter transcript

The model accounts for fines through transport and local transfers. Darcy provides the hydraulic response from the current resistance. The controlled comparison supports neglecting inertia for this regime, while permeability choice changes the prediction strongly. The next step is independent calibration and physical validation, rather than treating either illustrative result as measured truth. Our main validation input is measured pressure. PI remains useful for predictive machine-puck studies, and constant nine bar provides a controlled reference. Neither mode is universally preferable.

References and model documentation



- M. Bora and H. Briesen, “Characterization of Bimodal Particle Size Distribution of Ground Coffee Powder,” *Journal of Food Process Engineering*, 49, e70645, 2026. doi:10.1111/jfpe.70645
- S. Ergun, “Fluid Flow through Packed Columns,” *Chemical Engineering Progress*, 48, 89–94, 1952.
- FineFlow–Espresso technical report, Python solver, named-case configuration and automated verification suite accompanying this presentation.

Presenter transcript

The solver and technical report define the model. `compare_pressure_modes.py` saves the four matched constant-pressure and PI runs, with complete parameters, histories and fields. The separate `compare_closures.py` retains the frozen three-run PI sensitivity study. Constant pressure and Darcy are now defaults. PI, Forchheimer and alternative permeability closures remain available. Physical validation remains open.